

The Hedge India Doesn't Need to Build

India can stretch ethanol blending to E25–E27 this decade, and it should. Going beyond means building a second fuel system — a flex-fuel and E85 value chain — that Brazil began fifty years ago, before electric cars existed. We modelled it. Even phasing out every petrol car from next year cuts India's crude bill by only about 4% by 2030, almost all of it from blending and barely 1% from flex passenger vehicles. With EVs, hybrids and CNG arriving, the parallel supply chain is an expensive hedge whose moment has passed.

Thematic Equity Research — Indian Automotive & Energy Transition

Executive Summary

India crossed 20% ethanol blending in late 2025 and made E20 its standard petrol grade from April 2026 — a national target met six years early. The question now is whether the programme should climb higher: to E25 or E30 by blending, to E85 and flex-fuel vehicles beyond that, and whether the crude-import saving is worth the cost of building an entirely new fuel value chain to capture it. We discuss our answer in this report.

Our answer separates cleanly into two halves, and the separation is the whole argument. The first half — stretching the blend from E20 toward E25 or E27 over this decade — is achievable, sensible, and worth banking. The second half — going beyond the blend wall by building a national flex-fuel and E85 supply chain — is a fifty-year industrial project whose payoff, when you actually model it, is surprisingly small, increasingly duplicated by technologies arriving anyway, and justified mainly by a hedging logic that no longer fits India's moment. We are constructive on the first half and sceptical of the second.

Start with why the first half is the easy half. The climb from 1.5% to 20% blending was achieved by pouring a little ethanol into engines India already owned, on crops it already grew, asking nothing of the consumer. Even then it was not painless — official mileage loss at E20 is put at 1-6%, but a meaningful share of older-vehicle owners report closer to 10%, with engine and fuel-system complaints to match. Still, E20 worked because ethanol up to about 20% can live inside an ordinary petrol engine. That is the blend wall, and it is physics, not policy. New engines tuned for E20, mandatory since April 2025, lift the tolerable blend a little as the fleet turns over, which is why E25 and even E27 by 2030 is a credible base case — the specifications for E22 through E30 are notified and the excise duty on them waived.

But the wall does not move far. You cannot blend your way to E50 in a conventional engine, and the world confirms it: the United States, the largest ethanol producer on earth, has sat at E10 for fifteen years; Europe caps food-crop biofuel at 7% and bets on electrification. Blending, on its own, takes India to roughly E25–E27. Full stop.

Even that modest stretch rests on a foundation that moves with the weather, and this is the part of the story that gets least attention. India's ethanol does not come from a stable industrial input. It comes from the fields, and the fields depend on the monsoon. The programme learned this the hard way. When a poor monsoon in 2023 cut the sugar crop — yields in Maharashtra and Karnataka fell from 120–130 tonnes a hectare toward 80 — the government banned sugarcane juice for ethanol to protect the sugar bowl, and distilleries pivoted to maize. The consequence ran exactly counter to the programme's purpose. India, long a maize exporter of two to four million tonnes a year, became a net importer for the first time in a generation, buying nearly a million tonnes of corn from Myanmar and Ukraine in 2024, while maize-based ethanol became the costliest feedstock in the system. Read that carefully: to make a fuel meant to replace an oil import, India was importing grain. The forex saving leaked out the back door. In September 2025, with the monsoon recovered and sugar back in surplus, the government reversed the ban and cane flooded back. The system is neither permanently cane-led nor grain-led. It is monsoon-led — and the higher you set the blend, the more a single dry year threatens the whole structure.

Now the deeper question the second half raises, and the one that should anchor any investor's thinking: ethanol is not actually cheaper than petrol. At today's crude of roughly \$75 a barrel, a litre of crude costs India about ₹45, and a litre of petrol costs around ₹55–58 to make before tax. A litre of ethanol is procured at about ₹60–62 today, and rose above ₹71 during the grain pivot. Ethanol is dearer than the petrol it replaces; the cheapness consumers see at an E85 pump is entirely a tax-and-subsidy construct. So why go further at all? The honest answer — the one good argument for a flex-fuel future — is a hedge. Brazil's system lets drivers switch to ethanol when crude spikes and ethanol turns genuinely cheaper, and back to petrol when crude falls. During the recent war, when crude briefly topped \$100, a country with Brazil's fleet could have leaned on ethanol to blunt the shock. That option has real value, and we do not dismiss it.

But the hedge only pays in a crude spike, because only then does ethanol undercut petrol — and Brazil built the machine to exercise it over fifty years, starting with the oil shocks of the 1970s, when electric cars did not exist. India is contemplating building the same hedge now, and the calculus has changed. We built a model to size it. In an aggressive scenario — every petrol four-wheeler phased out of showrooms from next year, only flex-fuel cars sold — the slow turnover of the vehicle stock means that by the end of 2030 petrol vehicles are still 45–50% of the fleet, down only from today's 66–70%, while flex-fuel vehicles rise from zero to barely 10–12%. Even then, the total crude-import saving is about 4.3% — and roughly 80% of that comes from the blend rising toward E30 across the whole fleet, not from the

flex cars. The flex-fuel passenger-vehicle contribution by 2030 is about one per cent of the crude bill. Extend flex to two-wheelers and you add perhaps two points more, for around three per cent from flex in total, six to seven per cent all-in — a figure broadly similar to what EVs and hybrids in the four-wheeler fleet would deliver anyway, without anyone building a second fuel system.

That is the crux. The flex-fuel and E85 value chain — vehicles, a national network of E85 pumps, a dedicated high-blend feedstock supply — is a large, single-purpose investment whose distinctive contribution to India's crude bill, on our numbers, is on the order of one to three per cent by 2030. The world has run this experiment three times and watched the supply chain fail to repay the build. The United States never got E85 past 2% of stations; most flex owners did not know they had a flex car, and the fuel never beat petrol per mile. China abandoned a national E10 mandate the moment its grain surplus turned to deficit — precisely India's 2023 episode in advance. Thailand built E85 to scale on subsidy, then watched demand collapse 81% when the subsidy was cut, with EVs taking the baton; its retailers stopped selling E85 from February 2026. Only Brazil sustains high blends, and even Brazil, after half a century, runs an average of about E30.

So our conclusion holds the two halves apart. Bank the E25–E27 stretch: it is cheap, it uses what exists, and it rewards integrated, feedstock-flexible producers who can swing between cane and grain as the monsoon dictates. But think hard before committing the nation to a flex-fuel and E85 supply chain to chase the rest. It is a hedge against a crude spike, the stock turns over too slowly for it to matter this decade, the prize is a few per cent of the crude bill that blending and electrification largely deliver anyway, and it is justified by a 1970s logic — build your own fuel because there is no alternative to oil — that electric vehicles, hybrids and CNG have quietly overtaken. Brazil was right to build its hedge. India would be building it fifty years too late.

Investment Conclusion

Our central view is that India's ethanol programme should be underwritten in two distinct parts, and that the market conflates them. The first part — raising the blend from E20 toward E25–E27 across the existing fleet by 2030 — is achievable and bankable. Installed distillery capacity already exists, the September 2025 cane reinstatement has restored feedstock flexibility, the engine fleet is turning over to E20-tuned units, and the regulatory scaffolding (E22–E30 specifications, excise waivers) is in place. We are constructive here, with one standing caveat: the feedstock base is monsoon-dependent, so even E25 is, in part, a bet on the rain.

The second part — progressing beyond the blend wall through flex-fuel vehicles and a national E85/E100 supply chain — we view with caution that hardens the closer one looks at

the numbers. We built a stock-and-flow model of the vehicle parc. Its central finding is that vehicle stock turns over too slowly for aggressive flow-side policy to matter by 2030: even if every petrol four-wheeler were phased out of new sales from next year, petrol vehicles would still be 45-50% of the fleet at end-2030, and flex-fuel vehicles only 10-12%. The resulting crude-import saving is about 4.3%, of which roughly four-fifths comes from blending toward E30 and only about one percentage point from flex passenger vehicles. Including flex two-wheelers lifts the flex contribution to roughly three percentage points, and the all-in saving to six to seven per cent — comparable to what EV and hybrid penetration in four-wheelers would achieve regardless. The marginal return on building an entire parallel fuel system is, therefore, low.

The one substantive argument for building it anyway is the hedge: a flex fleet lets consumers switch to ethanol when crude spikes, as Brazil does. We take this seriously, but it is undercut on three counts. Ethanol is not cheaper than petrol pre-tax (roughly ₹61 against a petrol making-cost near ₹55), so the hedge pays only in a severe crude spike; Brazil justified the build over fifty years, beginning before electric vehicles existed; and EVs, hybrids and rising CNG penetration now offer import relief and decarbonisation without a single-purpose supply chain. Our posture: own the E25–E27 stretch through integrated, feedstock-flexible producers and the makers of E20 engines and strong hybrids; treat flex-fuel as a small, optional crude hedge rather than a national build-out, and price it accordingly. We would turn more constructive on the second part only on a structural crude price above \$90–100 per barrel, a genuine second-generation (non-food, non-irrigation) feedstock breakthrough, or a clear stall in the EV cost-and-range curve. Absent those, the parallel supply chain is capital chasing a single-digit prize.

1. India Has Finished the Easy Part — and the Real Question Is Whether to Build a Second Fuel System

The first thing to understand about India's ethanol programme is why it succeeded so fast, because the reasons are precisely the reasons the next leg is hard. Blending rose from under 2% a decade ago to 20% in 2025, ahead of every official timeline, and it did so on three favourable conditions. The blend went into engines India already owned, needing only minor changes to seals and rubber on new models. The feedstock came from crops India already grew. And the consumer was asked for nothing — no new pump, no price decision, no behaviour change. A programme that asks nothing of anyone scales easily. That is the story of the first twenty per cent.

It was not entirely costless, and it is worth being honest about the friction even at E20, because it foreshadows the resistance higher blends will meet. The government puts the mileage loss at E20 at 1-2% for vehicles designed for it and up to 6% for older ones.

Consumer surveys tell a less flattering story: a meaningful share of owners of pre-2023 vehicles report efficiency losses closer to 10%, alongside complaints of corrosion, harder cold starts, and fuel-system wear. These surveys are self-reported and confounded, and should not be read as engineering proof. But the political signal is real, and it matters, because every step up the blend ladder sharpens exactly these complaints.

Now the conditions reverse. Past the blend wall, ethanol needs a different engine — a flex-fuel vehicle the consumer must choose to buy. Past the easy feedstock, ethanol needs more cane from a water-stressed land or more grain from a food supply. And past E20, somebody must pay — the driver in a vehicle premium and a fuel choice, or the exchequer in a subsidy. The next leg asks something of everyone.

So the road forks, and the forks lead to very different places. One fork pushes a higher blend — E25, E30 — into the fleet India already owns and is buying. That fork, as the next two chapters show, dead-ends near E25 at the blend wall, and even getting there is a monsoon bet. The other fork builds an entirely new fuel system: flex-fuel vehicles, a national E85 retail network, and a dedicated feedstock supply to feed it. This second fork is open-ended in ambition but enormous in cost, and the central question of this report is whether the crude-import saving it delivers is worth what it takes to build. We answer that question with a model, in Chapter 6. The chapters before it establish what bounds the answer.

2. Into the Engines India Already Owns, Ethanol Stops Near E25 — and That Is the Ceiling of the Blending Fork

Take the first fork first, because the market assumes it runs further than it does. There is a physical ceiling to how much ethanol an ordinary petrol engine can burn, and it is not a matter of political will. Ethanol holds roughly 65% of petrol's energy by volume, absorbs water, and corrodes materials petrol leaves alone. At low blends these effects are trivial; as the blend climbs they compound — mileage falls, cold starts harden, fuel-system parts degrade in vehicles never built for them. The blend at which this becomes unacceptable for the legacy fleet is, by international consensus and India's own engineering, about 20%. This is the blend wall. It is why India planted its flag at E20 and not some higher number.

India does have a little room above the wall, and precision matters here. Since April 2025, new vehicles must carry E20-tuned engines, which exploit ethanol's high octane through higher compression to claw back part of the energy penalty. As the fleet turns over to these engines through the late 2020s, the tolerable blend creeps up. That is the genuine case for E25 and even E27 by 2030 — not a breakthrough, but a slow ratchet as old engines retire. The scaffolding is already in place: specifications for E22 through E30 are notified, and the excise duty on those grades waived. We treat E25 by 2030 as a credible base case and E27

as a reach.

But notice what this is and is not. It is a stretch of single percentage points, bought by replacing the vehicle fleet over years. It is not a ramp to E50 or E85, because the wall does not move that far. The clinching evidence sits abroad, and we take it up properly in Chapter 7; in brief, the United States produces more ethanol than any nation in history, has mandated it for two decades, and remains stuck at E10, because higher blends do not work in ordinary cars and forcing them is punishingly expensive. Europe caps food-crop biofuel at 7% by deliberate choice. Neither lacks ambition or ethanol. Both found the same wall.

The conclusion from the first fork bounds everything that follows. Blending into the fleet India owns and is buying reaches E25–E27 this decade and no further. As Exhibit 1 would show — tolerable blend against fleet vintage, with the legacy fleet capped near E20 and the post-2025 fleet stretching toward E27 — the ceiling is set by the cars on the road, not by the mandate on paper. Anyone promising India will blend its way to E50 is describing a fuel no ordinary engine can use. Past E30 is not a blending question. It is a vehicle question — the second fork — and before we cost it, we must deal with the constraint that limits even the first.

3. The Plants Are Built; the Feedstock Is Rain-Fed — Which Is Why Even E25 Is a Monsoon Bet

Even the modest stretch to E25 runs into a constraint the capacity figures hide. India does not lack distilleries. It lacks reliable feedstock — because its feedstock grows in fields that depend on the monsoon.

Begin with what is not the problem. India has built roughly 18–20 billion litres of installed ethanol capacity against E20 demand of about 11 billion litres, and the plants run at barely 25–30% utilisation. On paper there is room for E30 several times over, which is exactly why the distilling lobby pushes so hard for higher blends — it has built capacity that has nothing to do, and a higher mandate fills it. Capacity is not the binding constraint. It is a glut. As Exhibit 2 would show — installed capacity against ethanol actually required at E20, E25 and E30 — the plants can supply E30 today. The question is whether the fields can.

They can, in a good year. The real limit is sustainable feedstock — how much cane and grain India can divert to fuel without starving its sugar bowl, its food plate, or its water table. Independent estimates put that ceiling at roughly 6–10 billion litres; the government's optimistic figure is about 13.5 billion. The gap between those numbers is the gap between "E20 is already the limit" and "E25 has headroom," and the reason it is a range is that the answer changes every year with the rain.

Here is the episode that proves it, and it is the most important passage in this report. In 2023 a weak, El Niño monsoon — India's poorest rainfall in five years — hit the cane states hardest. Maharashtra and Karnataka, nearly half the national crop, saw yields collapse from 120-130 tonnes a hectare toward 80. With sugar suddenly scarce, the government did the politically obvious thing: in December 2023 it banned the diversion of sugarcane juice to ethanol, to keep sugar on shelves. Distilleries, needing feedstock, turned to maize. And the consequences ran exactly counter to the programme's purpose. Maize demand surged; prices climbed by roughly half; the poultry and feed industries were outbid by fuel. India, long a maize exporter of two to four million tonnes a year, became a net importer for the first time in a generation — buying nearly a million tonnes from Myanmar and Ukraine in 2024, a figure up almost eighty-fold on the year before, while its own exports collapsed. Maize-based ethanol became the single costliest feedstock at about ₹72 a litre, after the government raised its price 29% to attract supply.

Sit with what that means, because it punctures the headline rationale. Ethanol exists to swap an import — crude oil — for something grown at home. For two years, to make that ethanol, India was importing maize. It had traded an oil import for a grain import, and paid a premium for the privilege. The forex saving the programme is built to deliver was leaking out the back door. This was not a design flaw. It was what happens when the monsoon fails and a feedstock base built on annual crops buckles.

In September 2025 the government reversed the ban. The monsoon had recovered, two good years had refilled the reservoirs and the cane fields, sugar was back in surplus, and cane juice and molasses flooded back into the ethanol stream. Sugar-company shares jumped. The food-versus-fuel tension eased as fast as it had arrived. The system, in other words, is neither permanently grain-led nor permanently cane-led. It is monsoon-led — it tilts to cane when the rain is good and the sugar surplus large, and lurches to grain, with all the import leakage that entails, when the rain fails. As Exhibit 3 would show — feedstock mix by year against monsoon rainfall, with the 2023 ban and 2025 reversal marked — the feedstock follows the weather, not a plan.

This is why the cane reinstatement, welcome as it is, relocates the constraint rather than removing it. Cane is the cleanest feedstock per litre of ethanol, but it is among the thirstiest crops India grows, concentrated in precisely the states where water stress is most acute. Leaning the programme back onto cane eases the food fight and reopens the water one, and it leaves the whole edifice exposed to the next poor monsoon, which will come. You can stretch to E25 on this base in a good year. Betting E30 on it every year is betting on the rain — and, as the next chapters argue, building a permanent flex-fuel supply chain on it is betting the rain will cooperate for fifty years.

4. Ethanol Is Not Actually Cheaper Than Petrol — So the Whole Case for Going Further Rests on One Idea: a Hedge

Before costing the second fork, confront the fact that quietly undermines its economic premise. Ethanol is not cheaper than petrol. The entire popular case for blending — domestic fuel displacing expensive imported oil — assumes ethanol undercuts the petrol it replaces. At today's prices it does not.

Walk the numbers. Crude has settled at roughly \$75 a barrel since the mid-June 2026 ceasefire, down some 40% from its conflict peak above \$100. At ₹95 to the dollar, a litre of crude costs India about ₹45. A barrel yields many products, so petrol cannot be priced by simple division; but stripping the taxes out of the Delhi pump price of about ₹102 — excise of roughly ₹22, state VAT, dealer commission — leaves a petrol making-cost, refinery gate to forecourt, of around ₹55-58 a litre. Now set ethanol beside it. Ethanol is procured at about ₹60-62 a litre today on the cane-led mix, and rose above ₹71 during the grain pivot of 2024-25. The cheapest feedstock, C-heavy molasses, sits near ₹58; cane juice near ₹66; maize near ₹72. On every comparison that matters, ethanol costs as much as or more than the petrol it displaces, before a rupee of tax. The cheapness a consumer sees at an E85 pump is not the fuel being cheap. It is the tax being waived. It is a subsidy wearing the costume of a saving.

This forces the real question into the open. If ethanol is not cheaper, why push beyond the easy blend at all? Strip away the rhetoric and one serious answer remains, and it is a good one: a hedge. This is Brazil's genius, and it is worth understanding precisely. Brazil does not fix one blend and one price against the weather. It lets both float. The anhydrous blend in ordinary petrol sits in a band — historically 18 to 27%, now 30% — adjusted with the relative price of sugar and ethanol. More importantly, ninety per cent of Brazil's cars are flex-fuel, and at the pump the driver chooses between gasoline and pure ethanol on a simple rule: ethanol pays when priced below about 70% of gasoline, reflecting its lower energy. When crude spikes and petrol turns dear, ethanol crosses that threshold and drivers switch to it en masse, blunting the shock to the import bill. When crude falls, they switch back. The flex fleet is, in effect, a giant distributed hedge against oil-price spikes, exercised one tankful at a time.

The value of that hedge is not theoretical, and the last two months proved it. When the Strait of Hormuz crisis drove crude above \$100 in May and June 2026, India's pump prices climbed and its import bill swelled, with nothing to switch to. A country with Brazil's fleet and fuel network could have leaned on ethanol precisely when oil turned expensive — the moment a hedge is supposed to pay. We take this argument seriously; it is the strongest case for a flex-fuel future, and any honest analysis must grant it. The question the rest of this report asks is not whether the hedge has value. It is whether the hedge is worth what

India would have to spend to build it — and whether, in 2026, it is even the right hedge to build.

5. The Hedge Pays Only in a Spike — and Brazil Built It Over Fifty Years, Before Electric Cars Existed

Grant the hedge its due, then weigh it honestly, because two facts cut against building it now. The first is that the hedge pays only in a crude spike. Because ethanol is not cheaper than petrol at ordinary crude prices — the whole point of Chapter 4 — the flex driver switches to ethanol only when oil climbs high enough that petrol's making-cost finally exceeds ethanol's, and high enough to overcome ethanol's energy penalty at the pump. That is a real but occasional event: a tail-risk insurance, not an everyday saving. For most of any given decade, with crude in the \$60-80 range, an Indian flex fleet would run mostly on petrol, exactly as Brazil's does in cheap-oil years. You would have built an entire parallel fuel system to capture a benefit that materialises in the minority of years when oil spikes.

The second fact is about timing, and it is where Brazil instructs rather than tempts. Brazil began building this hedge in 1975, with Proálcool, in direct response to the oil shocks — at a moment when there was no alternative to oil for a car. There were no electric vehicles, no lithium-ion cost curve, no charging networks, no CNG retail to speak of. If you wanted to insure a transport system against oil, home-grown ethanol was the only instrument available, and Brazil spent fifty years and built a ninety-per-cent flex fleet and put E100 at every pump to make it work. The hedge made sense because it was the only hedge on offer.

That is no longer true, and it is the heart of our scepticism. India in 2026 has alternatives to oil that Brazil in 1975 could not imagine. Battery costs are falling year on year; the total cost of ownership of an electric car is converging on petrol and already favourable for two- and three-wheelers; ranges are extending and charging networks thickening every quarter. CNG penetration is rising across the passenger and commercial fleet. Strong hybrids cut petrol consumption 25-40% with no new fuel and no new infrastructure. Each of these reduces oil dependence — the very thing the ethanol hedge is for — and several of them decarbonise more durably as the grid greens. The opportunity cost of pouring national capital and policy attention into a single-purpose E85 supply chain is everything that same capital could do to accelerate the alternatives that hedge oil without needing a parallel fuel network at all.

So the question sharpens from "is the hedge valuable?" to "is this the hedge worth building, now, given what else is arriving?" Brazil answered yes in 1975 because it had to. India, in 2026, has options Brazil never had. As Exhibit 4 would show — the menu of oil-dependence hedges available to India by decade, from "ethanol only" in the 1970s to "EV, hybrid, CNG and ethanol" today — the case for a dedicated ethanol hedge weakens precisely as the

alternatives multiply. Before we conclude, though, we owe the reader the number that decides it: how much crude would the flex-fuel build actually save? That is what we modelled.

6. We Modelled It: Even Phasing Out Every Petrol Car From Next Year Barely Moves the Crude Bill by 2030

This is the calculation that should anchor the investment decision, and we built a stock-and-flow model of India's vehicle parc to make it. The model's logic is simple and its conclusion is hard to escape: vehicle stock turns over slowly, so even violent action on new-vehicle sales changes the fleet — and the fuel it burns — only gradually. The prize from the flex-fuel fork, captured honestly, is small.

Start with the most aggressive scenario we can responsibly model. Suppose that from next year India phases petrol four-wheelers out of showrooms entirely, and sells only flex-fuel cars from that point on. This is more drastic than any policy on the table. And yet, because the cars already on the road keep running for fifteen years, the stock barely moves by 2030. Our model puts the petrol share of the four-wheeler parc at still 45-50% at the end of 2030, down only from today's 66-70%, while flex-fuel vehicles rise from zero to just 10-12%. The other half of the fleet is electric, hybrid, CNG and the petrol cars already sold. The lesson is the one every fleet modeller learns: flows are not stock. You can transform what is sold and barely dent, within five years, what is driven.

Now attach fuel to that fleet and add an aggressive blend assumption — that India also pushes the petrol blend the whole way to E30 by 2030, for both cars and two-wheelers, which Chapter 2 already flags as a stretch. Run it through, and the total reduction in India's crude-import bill by the end of 2030 is about 4.3%. That number, on its own, sounds like a respectable return on the whole programme. But decompose it and the picture changes entirely. Roughly four-fifths of that 4.3% — about 3.4 points — comes from the blend rising toward E30 across the existing fleet. It does not come from flex-fuel vehicles at all. Only about one-fifth of the 4.3% — close to 0.9 of a percentage point, call it one per cent — is attributable to the flex-fuel cars. As Exhibit 5 would show — the 2030 crude saving split into its blending and flex-fuel components — the flex contribution is the thin slice, and the blend is the thick one.

Hold that result up to the light, because it reframes the entire debate. The flex-fuel and E85 value chain — the new vehicles, the national pump network, the dedicated feedstock — is the expensive half of the programme. The blend increase is the cheap half; it uses the fleet and the fuel system that already exist. And the model says the expensive half buys about one per cent of the crude bill from passenger vehicles by 2030, while the cheap half buys

more than three times as much. You would be building a second fuel system to capture the smaller share of the prize.

The picture improves only modestly if flex-fuel reaches two-wheelers, which are 61% of petrol demand and therefore the larger pool. Extend flex penetration into two-wheelers from next year and our model adds roughly two further percentage points of crude saving, lifting the flex-fuel contribution to around three per cent and the all-in 2030 saving — blending plus flex across both segments — to about six to seven per cent. And here is the sting in the tail: that six-to-seven per cent is broadly the same order of magnitude that EV and hybrid penetration in the four-wheeler fleet is on course to deliver over the same horizon anyway, through vehicles people are buying for their own economics, with no parallel fuel system required. The flex-fuel build, in other words, spends heavily to reach a destination the market is already walking toward by another road.

We are clear about the model's assumptions, because they drive the result and the reader should weigh them. It assumes new petrol four-wheeler sales fall to zero from next year, replaced by flex; it assumes the petrol blend climbs to a maximum-aggressive E30 by 2030 across both cars and two-wheelers; it builds the parc forward on current sales and scrappage rhythms with a roughly fifteen-year vehicle life; and it nets EV and hybrid penetration against the petrol pool. Soften any of these toward realism — a slower flex roll-out, a blend that stalls at E25, faster EV adoption — and the flex-fuel contribution shrinks further, not grows. The one per cent from passenger-vehicle flex is close to a best case for that channel by 2030, not a conservative one. The numbers here are modelled estimates, offered to size the decision, not to forecast it to the decimal.

The conclusion writes itself. Measured against the whole crude bill, the distinctive contribution of the flex-fuel and E85 supply chain — the part blending and electrification do not already deliver — is on the order of one to three per cent by 2030. That is a small return on a very large industrial undertaking. And the global record, which we turn to now, shows it is not a return that has reliably repaid the build anywhere outside Brazil.

7. The World Has Run This Experiment Three Times — and the Supply Chain Rarely Repaid the Build

The most important evidence on whether to build a high-blend supply chain comes from the countries that already did. Three ran the experiment in earnest, each failed in a different and instructive way, and each failure maps onto a specific Indian vulnerability. Only Brazil sustained it — and only on endowments and over a timescale India cannot replicate.

Take the United States first, because it had every advantage and still could not make the supply chain pay. America produces more ethanol than any nation in history and has

mandated it for two decades, yet E85 never escaped the margins. The failure was a stack of specific frictions, not a single cause. E85 reached barely 2% of fuelling stations — a chicken-and-egg trap in which retailers would not install pumps without flex drivers and drivers would not buy flex cars without pumps. Fuel-supply contracts actively obstructed E85, with clauses barring it from under the station canopy, imposing labelling and minimum-volume burdens, and threatening franchise termination. Most damning, a GM study found that around 70% of flex-fuel owners did not know their car could run on E85, and fewer than 10% ever used it — the vehicles existed only because they earned the automakers fuel-economy compliance credits, not because anyone drove them on ethanol. And underneath it all sat the economics: E85's 20-30% range penalty meant that even priced below petrol per gallon, it rarely beat petrol per mile. The conclusion of the world's largest ethanol industry, after twenty years, was that compelling higher blends costs around \$770 for every incremental gallon of ethanol delivered. India's 48 E85 outlets, its untested consumer awareness, and its E85 that currently costs more per kilometre than E20 are the same frictions at an earlier stage.

China is the cleanest parallel of all, because it failed exactly the way India's feedstock can fail. In 2017 Beijing announced a nationwide E10 mandate by 2020, conceived to digest a vast state corn surplus and clean the air. Within two years it was quietly abandoned. The surplus had vanished — drawn down by poor harvests and livestock demand — and the government concluded, in its own words, that any promotion of ethanol gasoline "must be based on the precondition that food security is guaranteed." The blend rate never exceeded about 2-4% against the 10% target. Scholars called it the impossible trinity of China's grain policy: a country cannot be self-sufficient in grain, improve its diet, and run an ethanol mandate at the same time. That is precisely the wall India hit in December 2023, when a poor monsoon forced it to choose between sugar on shelves and cane in fuel tanks — and chose sugar. A supply chain built to dispose of a surplus becomes a liability the moment the surplus turns to deficit, and in India the thing that turns surplus to deficit is the monsoon.

Thailand is the cautionary tale closest to what India is now attempting, because Thailand actually built E85 to commercial scale — and then dismantled it. Through generous Oil Fund subsidies that made E85 20-40% cheaper than petrol at the pump, Thailand stood up flex vehicles, E85 stations, and real consumer demand. It worked while the subsidy lasted. Then the State Oil Fund Act forced subsidies to be reined in; when the E85 subsidy was cut in late 2022, E85 consumption fell 81% in a single year, and E85 stations dwindled. Meanwhile electric-vehicle registrations multiplied roughly fivefold, taking over the decarbonisation job. By February 2026 Thailand's dominant retailer, PTT's OR, stopped selling E85 altogether, with a full phase-out of E85 and the older gasohol grades targeted for 2027 and the surplus ethanol — more than half of national capacity — redirected to sustainable aviation fuel. Thailand's E85 was a creature of subsidy that died when the subsidy did, with EVs inheriting its purpose. India's E85, priced below E20 only because tax is forgone on a fuel that costs

more to make, is the same creature; and India, too, has EVs arriving to inherit the purpose.

Against these three stands Brazil, the exception that proves the rule — and even Brazil should temper the ambition rather than inflame it. Brazil sustains high blends because it has what no one else has: sugarcane yields near 7,000 litres a hectare, an 8-10 times energy balance, abundant land and water, a pricing market that lets ethanol float against gasoline, and ninety per cent of its fleet already flex after a fifty-year build. And after all that, Brazil's effective average blend is about E30. The country most successful at this on earth, with every structural advantage and half a century of effort, runs roughly where India's blending fork can already reach without building a thing. As Exhibit 6 would show — high-blend outcomes against the years of build and the structural endowments behind them — Brazil is an outlier in every dimension India lacks, and its destination is one India can approach by the cheaper road.

8. Conclusion: Bank the Twenty-Five, Skip the Second Fuel System

India's ethanol programme deserves the credit it has earned. Reaching E20 across a vast and chaotic fleet, six years early, on a domestic feedstock, is a real achievement of policy and logistics, and the farm-income and energy-security benefits are not imaginary. We begin from respect for what has been done — and we separate, sharply, what should be done next.

Bank the blending stretch. E25, and with luck E27, by 2030 is achievable, cheap, and sensible. It uses the engines and the fuel system that already exist, it rewards the integrated, feedstock-flexible producers who can swing between cane and grain as the monsoon turns, and it asks little of the consumer beyond what E20 already does. The only caveat — and it is permanent — is that the feedstock is rain-fed, so even E25 carries a monsoon risk that no amount of installed capacity can engineer away. This is the bankable half of the programme, and we are constructive on it.

Think hard before building the rest. Progressing beyond the blend wall means constructing a second national fuel system — flex-fuel vehicles, an E85 retail network, a dedicated high-blend feedstock supply — and our model says the distinctive prize from that system is about one per cent of the crude bill from passenger vehicles by 2030, three per cent if it reaches two-wheelers, against an all-in six-to-seven per cent that blending and electrification largely deliver anyway. The case for building it rests almost entirely on a hedge: the Brazilian ability to switch to ethanol when crude spikes. That hedge is real, but it pays only in a spike, because ethanol is dearer than petrol in ordinary years; and Brazil justified the build over fifty years, beginning in the 1970s, when home-grown ethanol was the only way to insure a

car against oil. India in 2026 has other insurance — electric vehicles down the cost curve, hybrids that cut petrol use today, CNG spreading across the fleet — that hedges oil without a parallel supply chain and decarbonises more durably as the grid greens. The flex-fuel build is not wrong in principle. It is a hedge whose moment, for India, has largely passed.

The frame that holds the whole report together is the monsoon, and it is worth ending on. India has built an energy ambition on an agricultural base, and that base rises and falls with the rain. As long as the ambition stays modest — E20, soon E25, blended invisibly into engines that already exist — a rain-fed foundation holds, because a bad year can be patched, as 2023 showed. The higher the ambition climbs — E30, a flex fleet, billions in distillery and retail capital, a fuel network the country comes to depend on — the more a single dry monsoon threatens the whole structure, and the more the programme's own forex logic leaks when it is forced onto imported grain. Brazil built its hedge on land and water it had in abundance, at a time when it had no other choice. India would be building the same hedge on a rain-fed base it cannot count on, fifty years later, with better instruments already in hand.

For investors, the posture is therefore precise rather than directional. The value through this decade sits in the E25–E27 stretch and the feedstock-flexible producers who can feed it from whichever crop the monsoon favours, and in the makers of E20 engines and strong hybrids who win the fleet however the flex story plays. Flex-fuel is an option to hold cheaply and patiently, not a build to underwrite at scale — decided by levers like the E85 discount and GST parity rather than by economics as they stand. We would change this view on three signals: crude sustained above \$90-100 a barrel, which would make even costly ethanol macro-attractive and the hedge worth exercising; a genuine second-generation feedstock breakthrough that finally cuts ethanol free from the fields and the rain; or a clear stall in the EV cost-and-range curve that leaves ethanol as the only hedge standing. Absent those, the conclusion holds. India has blended the easy twenty, can reach a sensible twenty-five, and should think very hard before spending a fortune to build, half a century late, a hedge the future no longer requires.

Data and methodology notes: Petrol consumption ~40 MMT and total petroleum products ~239 MMT (PPAC, FY2024-25); diesel ~91 MMT; crude imports ~234 MMT at ~88% import dependence, ~\$137bn gross. Petrol segment split (two-wheelers ~61%, passenger vehicles ~34%, three-wheelers ~2-3%) per PPAC/Nielsen. Ethanol carries ~65% of petrol's volumetric energy. Crude ~\$75/bbl and INR ~95/USD as of mid-June 2026 (Brent ~\$79 following the US-Iran ceasefire, down ~40% from the May-June conflict peak above \$100); implied crude cost ~₹45/litre; petrol pre-tax making-cost ~₹55-58/litre derived from the Delhi retail price (~₹102.12) net of excise (~₹22), VAT and dealer commission. Ethanol procurement ~₹60-62/litre on the current cane-led mix; ~₹71.32 weighted average during

the 2024-25 grain pivot; C-heavy molasses ~₹57.97, cane juice ~₹65.61, maize ~₹71.86. Installed ethanol capacity ~18-20 billion litres at ~25-30% utilisation; E20 demand ~11 billion litres; sustainable feedstock ceiling ~6-10 billion litres (independent) vs ~13.5 billion (NITI Aayog 2021). Sugarcane concentration in Maharashtra/Karnataka/UP (>80% of output) and 2023 El Niño yield collapse (120-130 to ~80 t/ha) per ISMA/USDA FAS/Reuters; December 2023 cane-juice diversion ban and September 2025 reinstatement per government notifications. Maize a net import in 2024 (~0.9 MT, up ~7,940% YoY; exports down from 2-4 MT historically) per Ministry of Commerce/USDA FAS. Global cases: US E85 at ~2% of stations, ~70% of flex owners unaware (GM), ~\$770/incremental gallon (AFPM); China E10 mandate (2017) abandoned by 2019-20 on grain-deficit/food-security grounds, blend ~2-4% vs 10% target; Thailand E85 consumption -81% (2022-23) after subsidy cut, OR halting E85 sales 1 Feb 2026, full phase-out by 2027, surplus to SAF, per Nation Thailand/USDA FAS. Brazil E30 (Aug 2025), ~90% flex fleet, sugar-ethanol-linked blend band, ~70% pump-parity rule, ~7,000 L/ha, per CNPE/USDA FAS/IFPRI. THE VEHICLE-PARC MODEL: a stock-and-flow projection of India's four-wheeler and two-wheeler parcs to 2030 under an aggressive scenario — new petrol four-wheeler sales to zero from next year (flex-only), petrol blend to E30 by 2030 across both segments, ~15-year vehicle life with current sales/scrappage rhythms, EV/hybrid penetration netted against the petrol pool. Key outputs: 2030 petrol-vehicle stock ~45-50% (from ~66-70%); flex-fuel stock ~10-12%; total crude-import-bill reduction ~4.3%, of which ~80% (~3.4pp) from blending and ~20% (~0.9pp, ~1%) from flex passenger vehicles; flex in two-wheelers adds ~2pp (~3% flex total); all-in ~6-7%, comparable to expected EV+hybrid contribution in four-wheelers over the same horizon. All model figures are analyst estimates intended to size the decision, not point forecasts; softening any assumption toward realism reduces the flex-fuel contribution. Government benefit claims (forex saved, CO2 avoided, farmer income) are self-stated and not independently audited.